

Calculating Factorials

Many computer science courses use factorial calculation as a classic example of a recursive function. The factorial of an integer (let's call it n) is the product of all integers from 1 to n . For example, the factorial of 4 is $4 \times 3 \times 2 \times 1$, or 24. An exclamation mark is the math notation for factorials, as in $4!$, which means *the factorial of 4*. [Table 2-1](#) shows the first few factorials.

Table 2-1: *Factorials of the First Few Integers*

$n!$	Expanded form	Product
$1!$	$= 1$	$= 1$
$2!$	$= 1 \times 2$	$= 2$
$3!$	$= 1 \times 2 \times 3$	$= 6$
$4!$	$= 1 \times 2 \times 3 \times 4$	$= 24$
$5!$	$= 1 \times 2 \times 3 \times 4 \times 5$	$= 120$
$6!$	$= 1 \times 2 \times 3 \times 4 \times 5 \times 6$	$= 720$
$7!$	$= 1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7$	$= 5,040$
$8!$	$= 1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8$	$= 40,320$

Factorials are used in all sorts of calculations—for example, finding the number of permutations for something. If you want to know the number of ways that exist to order four people—Alice, Bob, Carol, and David—in a line, the answer is the factorial of 4. Four possible people can be first in line (4); then for each of those four options, three remaining people can be second in line (4×3); then two people can be third in line ($4 \times 3 \times 2$); and the last person left will be fourth in line ($4 \times 3 \times 2 \times 1$). The number of ways people can be ordered in line—that is, the number of permutations—is the factorial of the number of people.

Now let's examine both an iterative and a recursive approach to calculating factorials.

The Iterative Factorial Algorithm

Calculating factorials iteratively is fairly straightforward: multiply the integers 1 up to and including n in a loop. *Iterative* algorithms always use a